# Ahmad Amaan

*Platform Software Engineer 2, Oracle*

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## Education

**Indian Institute of Technology Kanpur** *2020 – 2024*
*B.Tech., Mechanical Engineering* CGPA: 8.83/10 | Minors: Computer Systems and Machine Learning

## Work Experience

**Platform Software Engineer 2, Oracle** *Oct 2025 – Present*

- Owned the end-to-end delivery of Seeded Workflow Requests across Commons, Deployment Service, and Job Scheduler, including validation, deployment processing, feature flags, persistence, integration testing, demos, and resolution of child-workflow failures.
- Built an AI-assisted remediation pipeline using custom OpenAI Codex skills to ingest Trivy scan results, remediate CVEs, verify project builds, and prepare merge requests with context-aware commit messages.
- Delivered the Temporal v1.28 uptake by implementing new persistence queries and validating it in pipeline and environment tests.
- Improved production reliability through on-call support, resolution of high-priority incidents, alert tuning, runbook updates, and Grafana dashboards covering 15+ operational metrics.

**Platform Software Engineer 1, Oracle** *Jun 2024 – Sep 2025*

- Added Oracle Database support to Temporal's open-source workflow engine by extending its SQL persistence layer, implementing Oracle-compatible schemas and queries, and adapting visibility logic across multiple releases.
- Built Temporal's CI/CD pipeline end-to-end for a multi-module Go codebase, adding 10+ stages across builds, artifact publishing, container image creation, and vulnerability scanning.
- Contributed to core microservices including Job Scheduler, Job Execution Manager, and Job Repository by implementing functional changes, building JAX-RS APIs, and adding unit and integration test coverage.

**Application Developer Intern, Oracle** *May 2023 – Jul 2023*

- Enabled SQL script invocation in the Spectra Schema Deploy Tool by integrating Oracle SQL*Plus into an init container workflow.
- Created and deployed Docker containers with Oracle Instant Client and Java 17 to enable SQL script execution within the microservice release workflow.

## Research Experience

**Visual Analysis of Prediction Uncertainty in Neural Networks for Deep Image Synthesis** *Aug 2023 – Nov 2023*
*Mentored by Prof. Soumya Dutta, IIT Kanpur*

- Conducted fine-grained pixel-wise analysis of epistemic uncertainty, absolute prediction error, and gradient-based sensitivity across RGB channels for MC-Dropout and Deep Ensemble methods.
- Evaluated view-conditioned deep image synthesis models on 3D scalar field datasets, quantifying uncertainty, robustness, and scalability across dropout rates and ensemble sizes.
- Co-authored a paper published in IEEE Transactions on Visualization and Computer Graphics (IEEE TVCG).

## Key Projects

**BorderPay** *Feb 2024 – Mar 2024*
*CS731 Endterm Project*

- Built smart contracts in Go on Hyperledger Fabric for identity, contract, and salary transaction management.
- Developed Node.js and Express APIs plus a React frontend with role-based dashboards for employers and employees.

**Fraud Detection** *May 2022 – Jul 2022*
*Project under Prof. Tushar Sandhan, IIT Kanpur*

- Processed 1.8 million imbalanced transaction records with 22 features and compared SVM, ANN, and Random Forest models using SMOTE-based resampling.
- Improved model quality through feature selection and preprocessing, achieving best recall of 0.93 with Random Forest.

**Codeforces Stress Testing Platform** *May 2022 – Jul 2022*
*Programming Club, IIT Kanpur*

- Built a concurrent online judge to stress test Codeforces problems with 95% accuracy on failing submissions.
- Implemented an auto-grader for GitHub submissions using Go, templating, concurrency, database integration, and web scraping.

## Skills

**Languages:** Java, Groovy, Go, Python, C, C++, TypeScript, SQL
**Frameworks & Libraries:** JAX-RS, Spock, Node.js, Express, React, Hyperledger Fabric, NumPy, Pandas, Matplotlib, Scikit-learn
**Tools:** Docker, Kubernetes, Helm, Git, Postman, Grafana, PromQL, Oracle SQL Developer, OpenLens, Jira

## Awards and Distinctions

- Earned promotion to IC2 at Oracle after receiving a 5/5 performance rating for strong technical contributions and delivery.
- Conferred Academic Excellence Award in 2022–2023 and 2023–2024 for outstanding academic performance by IIT Kanpur.
- Excellence in Computer Science Award for scoring 100 in Computer Science at both ICSE and ISC.